

the standard model of examination security assumes that custody of printed material can be audited after the fact through physical seals and human testimony alone this assumption fails precisely when the custodian is complicit because the seal reports only that the packet was opened not who photographed the contents once opened a forensic fingerprint embedded in the printed page itself shifts the evidentiary basis from testimony to signal each copy carries an invisible codeword drawn from a collusion resistant code so that any leaked photograph identifies the source unit with a provable bound on the probability of false accusation the marking survives the hostile channel of a phone camera and social media recompression because it is carried by low frequency geometry rather than fine texture the question this experiment answers is whether the physical channel preserves enough of that geometry to be useful at all which is the cheapest possible thing to falsify before any further investment in code design or learned carriers the standard model of examination security assumes that custody of printed material can be audited after the fact through physical seals and human testimony alone this assumption fails precisely when the custodian is complicit because the seal reports only that the packet was opened not who photographed the contents once opened a forensic fingerprint embedded in the printed page itself shifts the evidentiary basis from testimony to signal each copy carries an invisible codeword drawn from a collusion resistant code so that any leaked photograph identifies the source unit with a provable bound on the probability of false accusation the marking survives the hostile channel of a phone camera and social media recompression because it is carried by low frequency geometry rather than fine texture the question this experiment answers is whether the physical channel preserves enough of that geometry to be useful